



Research article

Environmental impact from vermicomposting of organic waste in Kampala, Uganda

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ARTICLE INFO

Article history:

Received 11 December 2015

Received in revised form

20 June 2016

Accepted 20 June 2016

Available online 5 August 2016

Keywords:

Manure

Sub-Saharan Africa

Emission factors

Lifecycle assessment

Fertiliser

Economic performance

ABSTRACT

Urban animal farming is becoming increasingly important in feeding the growing population of many sub-Saharan African cities. However, management of the animal manure generated is proving to be challenging due to space restrictions. Vermicomposting is one of the methods proposed to address this challenge. This study investigated the environmental performance of the vermicompost system by measuring the gaseous emissions generated from the system. In addition, the vermicompost system was compared with other manure management systems currently in use, using life cycle assessment (LCA) methodology. The emissions factors for the vermicompost system were found to be 10.8, 62.3 and 12.8 g/Megagram biowaste for methane, nitrous oxide and ammonia, respectively. LCA showed satisfactory performance of vermicomposting in terms of global warming and eutrophication potential, although if the vermicompost generated is dumped, this could lead to increased eutrophication. However, this is still much lower than the eutrophication caused by open dumping of untreated manure.

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1. Introduction

Economic development and the improvements in lifestyles witnessed in most low- and middle-income (LAMI) countries are being accompanied by generation of increased amounts of waste (Goorhuis, 2014). Management of municipal solid waste (MSW), especially organic waste, is a challenge with which authorities are grappling in many towns and cities in sub-Saharan Africa (SSA). According to Hoornweg and Bhada-Tata (2012), it is arguably the most important municipal service provided by municipal authorities and is usually a city's single largest budgetary item. It mainly involves the collection of waste and its transportation to landfill (Cheremisinoff, 2003; Zhu et al., 2008), with most of the budget being used for waste collection and the salaries of waste sweepers (Zhu et al., 2008). However in many SSA cities, less than half the

waste generated is collected and taken to landfill (OAG, 2010; Ofori-Boateng et al., 2013). Landfilling, like other end-of-pipe treatment methods and disposal methods, not only imposes high operating and capital costs, but is also associated with long-term liabilities (Cheremisinoff, 2003). These include acting as a source of leachate and other contaminants that can pollute underground aquifers and surface water. Landfills are also recognised as a major source of anthropogenic methane emissions and an important contributor to global warming, accounting for up to 19% of methane emissions in the world (Kumar et al., 2004). In addition, high land prices as a result of rising populations and incomes in many cities (Idris et al., 2004) have greatly reduced the economic attractiveness of landfilling, thus fostering initiatives to divert waste prior to landfilling. Finally, wastes also contain resources that could meet the demand for energy and vital plant nutrients (Komakech et al., 2014b) and all these resources are lost when the wastes are landfilled.

Because of the pronounced disadvantages of landfill, countries worldwide have introduced various measures to limit its use. In many high-income countries, only inert waste is accepted in landfill and other waste is subjected to different waste treatment methods, such as composting, recycling and incineration with energy recovery (Geng et al., 2010). In LAMI countries, measures to reduce

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the waste going to landfill mainly involve value addition to the dominant organic waste stream. These include anaerobic digestion (AD), composting and incineration with energy recovery, with composting being the most prominent (Hoornweg and Bhada-Tata, 2012; Zurbrugg et al., 2005). However these efforts have not been particularly successful (Oteng-Ababio et al., 2013; Parawira, 2009). In the case of composting, for instance, one of the major reasons for its failure is the lack of a ready market for the fertiliser produced (Ngoc and Schnitzer, 2009), thus showing how unappreciative farmers are of this organic waste value chain. Compared with chemical fertilisers, bio-fertilisers have the disadvantage of diluted nutrients. The most used fertiliser globally, urea, contains 46% nitrogen, whereas manure contains only 0.5% and the nitrogen is also less-plant available than that in urea. This results in a major difference in the field work required to apply 1 kg of nitrogen (2.2 kg urea versus 200 kg of compost). For AD, according to Parawira (2009), economic, political and technical factors present the major challenges to its development. There is therefore an urgent need to investigate the attractiveness of other organic waste value addition technologies.

One such technology is vermicomposting, which is the process by which worms are used to convert organic wastes into a humus-like substance called vermicompost (Munroe, 2007). Another by-product of this process is the worm biomass, which can be used as a suitable source of animal protein (Lalander et al., 2015). In previous experiments carried out in Kampala on non-optimised vermicompost reactors, it was established that vermicomposting performs well, with a material reduction rate of 45.9%, a waste-to-biomass conversion rate of 3.5% and a return on investment of 275% when treating 450 kg cow manure. However, the experiments also established that the vermicompost reactors were associated with poor performance in terms of pathogen reduction (Lalander et al., 2015). Data on greenhouse gases (GHG) and other emissions from vermicompost reactors in SSA are not readily available. Thus the objective of the present study was to measure emissions of GHG and other emissions from vermicomposting of mainly animal wastes in Kampala, Uganda, and to compare the results to emissions from waste management practices currently in use for managing such waste in Uganda, namely open dumping and storage for use as fertiliser. The overall aim was to test whether adoption of vermicomposting of waste could provide a reduction in GHG and other emissions associated with current waste management. In addition, a life cycle assessment (LCA) and economic comparison studies were performed to compare the environmental and economic performance, respectively, of vermicompost and of other manure management systems.

2. Materials and methods

Direct emissions were measured from small-scale vermicompost reactor units set up at Makerere University Agricultural Research Institute Kabanyolo (MUARIK). The LCA methodology was used to perform the environmental comparison and evaluate the environmental improvement of the vermicompost technology and the current baseline technologies used to handle the waste. The methods used are discussed further in the following section.

2.1. Study area

The vermicompost units were established in Makerere University Agricultural Institute Kabanyolo (MUARIK), which is located about 21 km north of Kampala city at coordinates 0°27'03.0" N and 32°36'42.0" E (Komakech et al., 2015b). The area receives about 1300 mm of rainfall per annum, while its temperature varies between 20 and 28 °C. MUARIK has a dairy farm, which was the

source of the animal manure used in the vermicompost reactors. The operation and other details of the vermicompost reactor have been described previously (Lalander et al., 2015). The reactors were made of hardwood pallets (three in number), measuring 1 m by 1 m by 0.8 m, each filled with manure. Indigenous earthworms (species *E. eugeniae*) picked from manure piles were used. Worm density at the time of measurement was about 4 kg/m².

2.2. Measurements of emissions

The vermicompost units were wrapped in high-density poly-ethylene film and sealed with cello tape to minimise gaseous losses. The gas in each of the three vermicompost reactors was sampled on four occasions in April 2014, details of which are shown in Table 1. At each sampling, gas was collected from the sealed vermicompost using four 60-mL gas syringes, which were then tightly sealed, suitably packaged for transportation as shown in Fig. 1 and taken to the government analytical laboratory for measurement of nitrous oxide, ammonia and methane. Nitrous oxide was analysed using a gas chromatograph (PerkinElmer Clarus 680 with electron capture detector and temperature range 100–450 °C), following procedures specified by Ermolaev et al. (2014), while ammonia was measured using a Gas Alert Max Xt II multigas detector (BW Technologies, Schaumburg, Illinois, USA) fitted with an ammonia sensor. Methane was measured using a Crowcon Triple Plus IR mode gas detector (Bristol, UK). The data collected were analysed using two-way ANOVA and Tukey tests in R statistical software to check for differences in gas composition. The procedures used were as specified by Venables et al. (2012). However, the airflow velocity from the vermicompost reactor could not be measured because the flow rates were too low. To overcome this problem, emissions factors were estimated based on the dynamic respiration index (DRI) as proposed by Lleó et al. (2013). The DRI value used was determined from data presented by Lalander et al. (2015).

2.3. LCA methodology

The LCA methodology used to perform the environmental comparison between vermicompost technology and the current methods being used to handle the waste is explained in more detail in the following sub-sections.

2.3.1. Goal, scope and functional unit

The goal of the study was to evaluate the vermicompost method of treating animal manure (Scenario Ib) in Kampala in terms of the environmental impacts (global warming potential (GWP) and eutrophication potential). This was compared with the current ubiquitous practice of dumping animal manure (Scenario Ic: Baseline condition) or using it as fertiliser in gardens (Scenario Ia) (Komakech et al., 2014a). The LCA performed also sought to identify

Table 1
Sequence of measurement of emissions.

Time	Activity
8.00	Cover vermicompost units with plastic
9.00	Sample gas in the units and uncover units ^a
10.00	Cover vermicompost units with plastic
11.00	Sample gas in the units and uncover units
12.00	Feed the units
13.00	Cover vermicompost units with plastic
14.00	Sample gas in the units and uncover units
15.00	Leave units uncovered
16.00	Cover vermicompost units with plastic
17.00	Sample gas in the units and uncover units

^a To ensure that the vermicompost units were not starved of oxygen supply.

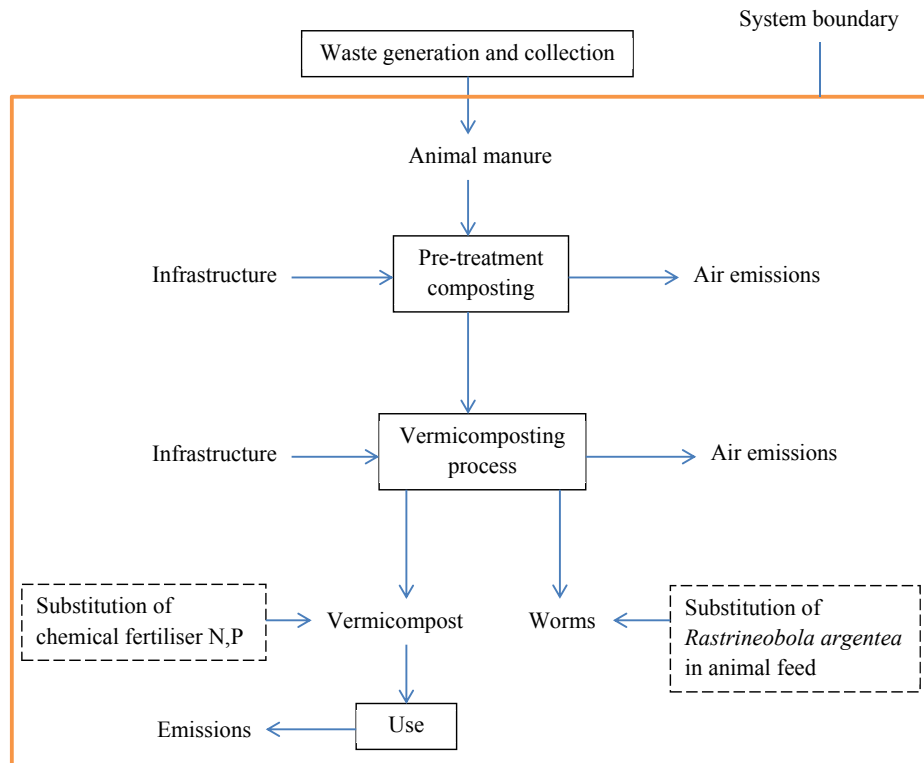


Fig. 1. Schematic diagram showing the system boundary of the vermicomposting system (Scenario 1b).

processes contributing most to the environmental impacts associated with a particular practice, so that possible improvements in the life cycle of the different technologies could be suggested.

The functional unit used was one tonne of impurity-free animal waste treated to produce a quality soil improver/fertiliser.

2.3.2. System boundaries

The system boundaries used in the study are shown in Figs. 2–4. The system analysis started with the treatment system receiving animal manure waste at the gate of its pre-treatment section and ended after the soil improver or fertiliser had been applied in the

field. Waste generation, collection and sorting processes prior to treatment were neglected because they were the same for the different treatment technologies. The study was designed for Kampala as a case study, but when data were lacking, values for other cities were used. A more detailed description of the different animal waste treatment/handling technologies is given below.

2.3.2.1. Vermicompost system. The vermicompost system at MUARIK as described by Lalander et al. (2015) was used. A schematic diagram of this system is shown in Fig. 1. Fresh animal manure was received and stored at the pre-treatment section for

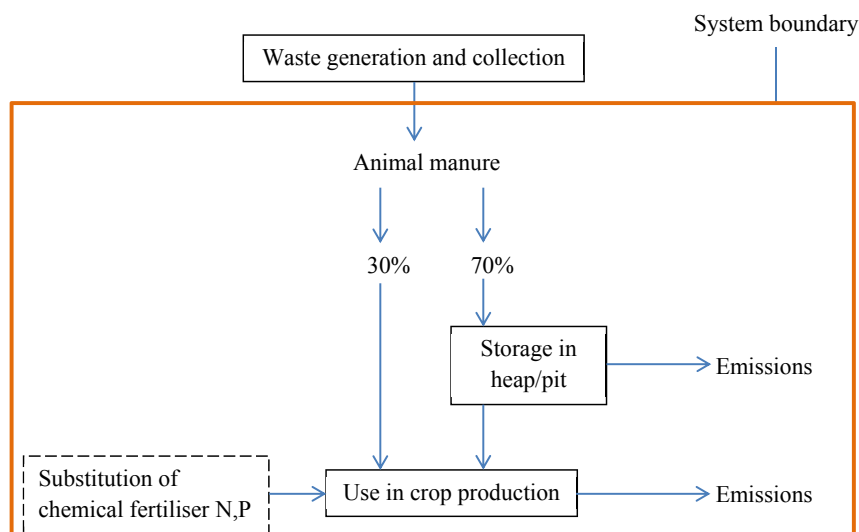


Fig. 2. Schematic diagram showing the system boundary of the manure into fertiliser system (Scenario 1a).

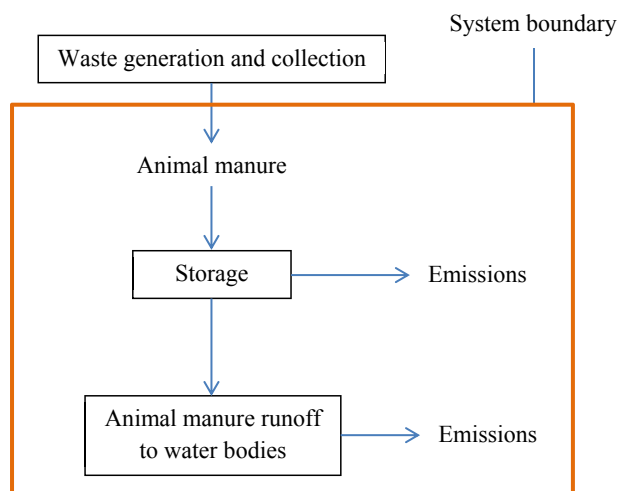


Fig. 3. Schematic diagram showing the system boundary of the baseline system (Scenario 1c).

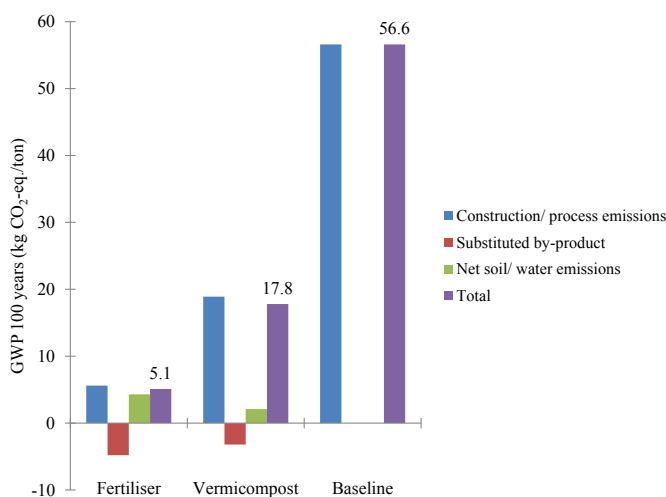


Fig. 4. Global warming potential (GWP) [kg CO₂-eq. ton⁻¹] of the different manure management systems.

four days. During this process, the manure undergoes some composting. This helps to reduce the concentration of ammonia, which would otherwise be harmful to the worms. The pre-treated manure was then fed into the vermicompost reactor, where the worms converted it into vermicompost and worm biomass over three days. The former was considered to replace mineral fertiliser and the latter to replace silver cyprinid, which is the main protein source in the production of animal feed in Uganda. The vermicompost was then assumed to be transported manually to the field for crop production.

2.3.2.2. Use of manure as fertiliser (fertiliser system). According to Komakech et al. (2014a), 32% of the farmers in Kampala use animal manure as fertiliser in their crop fields. Of this number, 70% prefer to store the fertiliser in a pit/heap before applying it in the field, while the remaining 30% apply the manure directly in the field without storage. Transport to and application in the field is carried out manually (Komakech et al., 2014a). Storage of manure is for two months (Zake et al., 2010). The manure is assumed to replace mineral fertiliser. A schematic diagram of this system is shown in Fig. 2.

2.3.2.3. Baseline system. According to Komakech et al. (2014a), 60% of the animal manure generated in Kampala is openly dumped as a result of low level of awareness among urban farmers in Kampala. Banadda et al. (2009) report that this manure, when carried away during rainfall events, finds its way into Lake Victoria, located about 8 km from the centre of Kampala, contributing to its pollution. In this study, due to the frequent rainfall received by the study area, an assumption was made that the dumped manure is subjected to aerobic degradation for one week before being carried by runoff rainwater to nearby water bodies (see schematic diagram in Fig. 3).

2.3.3. Impact categories

The impact categories considered in this study were emissions causing potential global warming and potential eutrophication. These factors were chosen on the basis of environmental relevance and availability of data. The characterisation factors used were taken from Solomon et al. (2007) for global warming (1, 25 and 298 CO₂ equivalents (CO₂-eq.) for carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O), respectively) and from CML (2002) for eutrophication (0.13, 0.35 and 0.10 phosphate-equivalents (PO₄³⁻-eq.) for nitrogen oxides (NO_x), ammonia (NH₃) and nitrate (NO₃), respectively).

2.3.4. Inventory

The animal waste received at the treatment unit was assumed to be similar to that treated by the vermicompost system described by Lalander et al. (2015), with 18.8% total solids (TS) and 14.7% volatile solids (VS), i.e. 72% of TS was VS. The amounts of nitrogen (N) and phosphorus (P) in the pre-treated animal waste were also assumed to be similar to those specified by Lalander et al. (2015) as 3.07% N and 0.046% P, while the carbon (C) was assumed to be 51% of VS (Jönsson et al., 2005). A brief description of other major items/activities used in the study is given in Table 2 (Supplementary materials).

The organic fertilisers produced by the different waste treatment systems contain nutrients that can substitute for industrial mineral fertiliser. The amount of fertiliser substituted (Mass_{subs}) depends on the content of nutrients in the organic fertiliser (Input_{nutrients}) and the substitution efficiency (Sub_{eff}) (Boldrin et al., 2009). This was calculated using Equation (1) with Sub_{eff} = 100% for P and 30% for N (Hansen et al., 2006).

$$\text{Mass}_{\text{subs}} = \text{Input}_{\text{nutrients}} \times \text{Sub}_{\text{eff}} \quad (1)$$

The energy and emissions saved through avoiding the production process of mineral fertiliser were taken from Brentrup and Pallière (2008) and Davis and Haglund (1999), respectively (Table 3).

The emissions from the transportation process were calculated using the Network for Transport Measures (NTM) freight calculator (Table 2), as specified by Komakech et al. (2015a).

The C in the organic fertiliser produced can be characterised as partly degradable and partly stable. The latter is stable C that can be bound for long periods and can be credited with avoided CO₂ downstream emissions (Favoino and Hogg, 2008). The amount of CO₂ bound (CO₂ sink) can be calculated as specified by Komakech et al. (2015a). However, for warmer climates like those of SSA, the value of the stable C is very low (Nyamangara et al., 2004) and for this study it was assumed to be zero.

The application of treated organic fertiliser to land can result in emissions of N₂O in an amount which can be calculated from IPCC (2006b) for the fertiliser system. For the vermicompost system, according to Boldrin et al. (2009) the value is given by Equation (2):

$$\text{N}_2\text{O} = \text{N}_{\text{input}} \times \text{N}_2\text{O}_{\%N} \times 44/28 \quad (2)$$

Table 2

Energy and emissions from the manufacturing process for mineral fertiliser.

Nutrient	Energy (MJ/kg)	Total GHG (g/kg CO ₂ -eq)	Other emissions (mg/kg)			
			NH ₃	PO ₄ ³	SO ₂	NO _x
N	29.8	2740	0.06	0.56	4390	4680
P	30.5	1670	1460	0.36	38,300	18,300

where N_{input} is the N content of the organic fertiliser and $N_{2O\%N}$ is the fraction of nitrogen converted to N_2O . Boldrin et al. (2009) reported that this value varies between 1.0 and 2.2%. For this study, a value of 1.6% was used. In the case of industrial mineral fertiliser, a value of 1.0% (IPCC, 2006b) was used.

Further losses of N from the organic fertiliser can occur in the form of nitrate leachate and runoff to groundwater and surface water, respectively. According to Hansen et al. (2006), the equations for determining nitrate leaching and runoff are:

$$NO_{3(gw)} = K_{NO3(gw)} \times N_{tot} \times 62/14 \text{ (for nitrate leaching to groundwater)} \quad (3)$$

$$NO_{3(sw)} = K_{NO3(sw)} \times N_{tot} \times 62/14 \text{ (for nitrate runoff to surface water)} \quad (4)$$

The coefficients used for surface water and groundwater were 0.08 and 0.07, respectively, for the vermicompost and fertiliser systems, as specified in Hansen et al. (2006) for loam soil. In the case of industrial mineral fertilisers, loss of nitrates through leaching and runoff was assumed to be negligible due to the high evapotranspiration rates experienced in SSA countries (during the vegetative period when the fertiliser is applied) (Brentrup et al., 2000).

Ammonia volatilisation losses were calculated as specified in Komakech et al. (2015a). The nitrogen in manure that ends up in water bodies was assumed to be transformed into nitrates (50%) and ammonia (50%).

Avoided emissions when worms substituted for silver cyprinid fish in animal feed were determined following the methodology used by Komakech et al. (2015a).

3. Results and discussion

3.1. Emissions from small-scale vermicompost reactors

The emissions measured from the vermicompost unit are shown in Table 4.

Two-way ANOVA test showed no significant differences ($P > 0.05$) in the gas concentration between the different sampling times. However, the same test showed a significant difference ($P < 0.05$) in concentration between the different gases. Further analysis by Tukey tests revealed that the concentration of nitrous oxide was significantly higher ($P < 0.05$) than that of methane and ammonia. However, the same test showed no significant difference ($P > 0.05$) between the concentration of methane and that of ammonia. This finding of higher nitrous oxide emissions from the vermicompost reactor is similar to results reported by Frederickson and Howell (2003), who found high emissions of nitrous oxide from unheated vermicompost beds during summer. Moreover, the emissions were positively correlated with worm density, as earthworms are suspected to be high emitters of nitrous oxide (Frederickson and Howell, 2003). The worm density in the vermicompost reactor used in the present study was very high, as previously reported by Lalander et al. (2015). It is therefore probable that the high nitrous oxide emissions noted could be a result of high

Table 3

Emissions measured from the vermicompost unit at the different sampling times.

Sampling no.	Content (percentage of total gases)		
	CH ₄	N ₂ O	NH ₃
1	0.16	0.33	0.12
1	0.14	0.28	0.16
1	0.09	0.27	0.07
1	0.17	0.28	0.26
2	0.12	0.25	0.14
2	0.12	0.30	0.12
2	0.19	0.28	0.08
2	0.10	0.26	0.07
3	0.09	0.29	0.15
3	0.20	0.26	0.25
3	0.20	0.34	0.14
3	0.16	0.27	0.25
4	0.13	0.33	0.17
4	0.10	0.28	0.06
4	0.15	0.30	0.25
4	0.08	0.30	0.16
Mean ± standard deviation			
Emissions factor (g Mg ⁻¹)			
10.80 ± 0.04 62.30 ± 0.03 12.80 ± 0.07			

worm density.

The emissions factors for vermicomposting could not be determined because the velocity of gases emitted could not be measured owing to low airflow rates. However, the emissions factors were estimated based on the DRI value (Lleó et al., 2013). The airflow was estimated to be $0.0005 \text{ m}^3 \text{ kg}^{-1} \text{ biowaste day}^{-1}$. On the basis of this and taking into account the gaseous concentrations, the emission factors for small-scale vermicompost system were calculated as 10.8, 62.3 and $12.8 \text{ g Mg}^{-1} \text{ biowaste}$ for methane, nitrous oxide and ammonia, respectively. The values obtained for methane and nitrous oxide were generally higher than those reported by Lleó et al. (2013). The higher values of methane are probably an indicator of anaerobic conditions in the vermicompost reactor, which according to Lalander et al. (2015) arise when there is excessive loading of the reactor. The high ammonia and nitrous oxide values may also be attributable to the higher nitrogen content of the substrate (Lalander et al., 2015). Nevertheless, when compared with compost systems (Cuhls et al., 2011; Lleó et al., 2013), emissions from the vermicompost system were much smaller, thus representing an effective method to minimise emissions from waste treatment.

3.2. Mass balance of nutrients from alternative manure management systems

The nutrient content (C, N, P) of the end product of the different manure management systems is shown in Table 5.

The C in the compost of the baseline and fertiliser systems was calculated by subtracting C lost in methane and carbon dioxide (gas), using the values specified in Table 2. Data on the C in the compost, worms and gas of the vermicompost system were taken from Lalander et al. (2015). The N and P values for the incoming and outgoing waste were derived from Lalander et al. (2015). The N amount leaving the baseline and fertiliser systems was less N losses

Table 4
Changes in mass balance of nutrients for the different manure management systems.

Element (kg tonne ⁻¹ of wet manure)	Nutrients out (kg tonne ⁻¹ wet manure waste)								
	Baseline system			Vermicompost system			Fertiliser system		
	Compost	Gas	Leachate	Compost	Worms	Gas	Compost	Gas	Leachate
C	74.97	74.06	0.91	<0.01	30.94	3.3	40.7	68.53	6.44
N	5.77	4.86	0.88	0.03	2.55	0.63	2.59	4.81	0.80
P	0.05	0.05	<0.01	<0.01	0.05	<0.01	<0.01	0.05	<0.01

in direct process emissions (Table 2) while the P was unchanged. Although Sommer and Dahl (1999) established that 6.3% of P is lost through leaching and other unknown sources, in this study it was considered to be negligible.

3.3. Sensitivity analysis

Three sensitivity analysis were performed. The first investigated the effect of having 100% of farmers applying the manure directly in the field without storage (Scenario II). The second investigated the effect of having 100% of the farmers storing the fertiliser before application in the field (Scenario III). The third investigated the effect of all the vermicompost produced being dumped without any application in the field, a scenario common among farmers whose interest is in worms as an animal feed (Scenario IV). These were compared with the baseline conditions (Scenario I).

3.4. Economic comparison

In an attempt to understand the widespread dumping of animal waste in Kampala as opposed to transportation and application to crop fields, the cost of transportation and application of 1 kg N in the different organic fertilisers was compared. For this, cost values in and around Kampala city for 2015 were used as follows: USD 40 for hiring a 4-tonne truck per trip, USD 4 for loading the truck and USD 40 for spreading the organic fertiliser on 1 ha of land. The assumption made was that the organic fertilisers would be transported from the animal farm to crop fields 15 km away. The amount of organic fertiliser required was calculated as the amount that could supply an application rate of 90 kg N per hectare (Bayite-Kasule, 2009), with the knowledge that only 30% of nitrogen supplied by organic fertilisers is plant-available (Hansen et al., 2006). This cost was compared with the cost of 1 kg N from mineral fertiliser based on 2015 prices.

3.5. Summary of material inputs and emissions for the different management systems

A summary of material inputs and emissions for the different biodegradable waste treatment technologies is provided in Table 6 (Supplementary materials), Fig. 4 (Global warming potential) and Fig. 5 (Eutrophication potential).

As Fig. 4 shows, the GWP was highest in the baseline system, followed by the vermicompost system and the fertiliser system. The baseline system's poor performance in terms of GWP is mainly due to the huge emissions of methane and nitrous oxide generated due to the dumping of manure (Table 6). This finding is in agreement with other studies, which report that open dumping of waste could be associated with a GWP as high as 1000 kg CO₂-eq. tonne⁻¹, a value that was much higher than for other alternative waste management methods considered (Manfredi et al., 2009). Our results also show that the vermicompost system performed poorly compared with the fertiliser system. This was mainly due to the higher process emissions of nitrous oxide (Table 6), which is

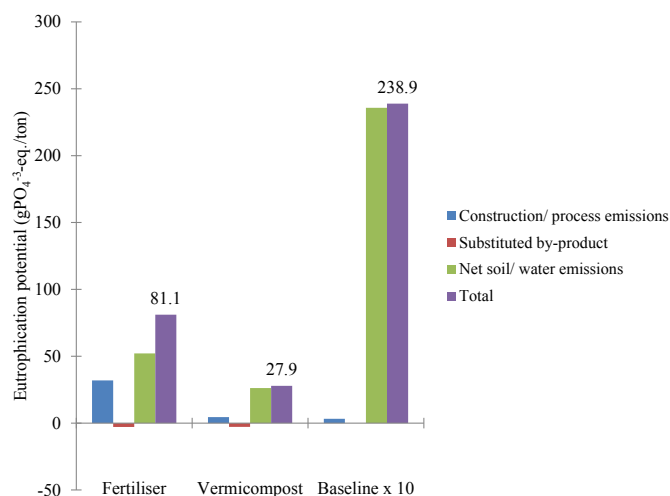


Fig. 5. Eutrophication potential [g PO₄³⁻-eq. ton⁻¹] of the different manure management systems.

probably emitted by the worms (Frederickson and Howell, 2003). It is interesting to note that the GWP for vermicompost obtained in this study was similar to that obtained in a previous study by Komakech et al. (2015a), even though the process emissions used in the studies were different. Thus optimisation of large-scale vermicompost systems as regards GWP may require analysis of the different parts of the system and not only the direct process alone. The fertiliser system performed best mainly because of the lower process emissions associated with it, which translated into higher plant nutrient availability (Table 5). If stakeholders are more interested in use of manure to replenish plant nutrients in their crop fields, then this system should probably be considered. However, this attribute of the fertiliser system should be treated with caution, as according to Atiyeh et al. (2000) vermicompost has a higher concentration of nitrates, which is the more plant-available form of nitrogen. In addition, as pointed out by Lalander et al. (2015), the vermicompost system is associated with the production of both compost and biomass (animal protein). The latter is especially important since in many African countries, including Uganda, there is an acute shortage of animal feeds and the few available on the market are highly contaminated, causing loss of animal productivity (Kagolo, 2012). Thus the vermicompost system has the ability to make a contribution towards addressing this anomaly.

A graphical representation of the eutrophication potential of the different manure management systems analysed in the present study revealed a slightly differing pattern (Fig. 5).

Again, the baseline system was by far the worst performing system. This was mainly because of the huge quantities of nutrients it delivers to adjacent water bodies. Banadda et al. (2009) report that Lake Victoria is experiencing a high level of eutrophication due to anthropogenic activities. Based on the results of the present

Table 5
Sensitivity analysis on global warming potential (GWP) and eutrophication in Scenarios I–IV.

Scenario	Environmental performance of different systems under different scenarios (per tonne of waste)		
	System	GWP (kg fossil CO ₂ -eq.)	Eutrophication (g PO ₄ ³⁻ -eq.)
I: Baseline	a. Fertiliser	5.0	81.1
	b. Vermicomposting	17.8	27.9
	c. Baseline	56.6	23900.0
II: 100% direct application of manure	Fertiliser	−0.4	86.0
III: 100% storage of manure	Fertiliser	7.4	78.1
IV: Vermicompost dumped	Vermicomposting	18.6	1240.0

Table 6
Economic comparison of different manure management systems.

Manure management system	N content	TS	N in one ton (kg)	Tonnage required for one ha	No. of trips	Total cost (USD)	Cost per kg N (USD)
Fresh manure (fertiliser)	3.1	18.8	5.8	52.0	13	612	6.8
Vermicompost	2.5	28.2	7.2	41.9	10	480	5.3
Manure (stored)	2.8	18.2	5.1	59.1	15	700	7.8
Vermicompost (dumped)	2.5	28.2	7.2	41.9	10	40	—
Fresh (dumped)	3.1	18.8	5.8	52.0	13	52	—

study, it is probable that the rampant dumping of animal manure (Komakech et al., 2014a) could in some way also be contributing to the increased levels of eutrophication of Lake Victoria. However, further studies are required to determine the amount of eutrophication of the lake that can be attributed to dumping of animal manure.

The fertiliser system was the next worst performing system. This was mainly as a result of emissions of ammonia from the process and from fertiliser application (Table 6). According to Sommer and Dahl (1999), the different ways of treating animal manure affect the amount of ammonia emitted, with higher emissions associated with compression and mixing of the animal manure compared with leaving it untreated. In this study we used the prevailing manure management conditions of Kampala, where stored animal manure is left untreated. Thus the ammonia process emissions should be considered as the bare minimum and, in the case of use of animal manure as a fertiliser, the current manure management practice of not treating the stored manure should probably be maintained.

The vermicompost system was the best performing, mainly due to lower ammonia emissions from the process and from application. Compared with previous values reported by Komakech et al. (2015a), the eutrophication potential of the vermicompost system in this study was six-fold lower. This was mainly due to the lower ammonia process emissions in this study compared with the previous study. According to Webb et al. (2012), factors such as increased air velocity, air temperature, surface area and pH lead to increased ammonia emissions. Thus to reduce the ammonia emissions, these factors have to be regulated.

Sensitivity analysis showed that direct application of animal manure to the field (Scenario II) had a great impact on GWP compared with the current baseline system (Scenario I). On the other hand, storage of the manure for at least two months before application in the field (Scenario II) worsened the GWP. However, these different changes had a negligible effect on the eutrophication potential (Table 7). When vermicompost was assumed to be dumped (Scenario IV), both the GWP and eutrophication potential were worse than if it had been used as a fertiliser (Scenario I). Therefore from an environmental perspective, it is not advisable to dump the vermicompost. However, by introducing vermicomposting a manure value chain is introduced where the animal feed can be sold, giving an incentive for management of the manure. If the manure is still dumped after treatment, the GHG and

eutrophying emissions from the manure are one third and almost half, respectively, of those from manure dumped directly.

As can be seen from Table 8, the economic comparison revealed that the most expensive manure management system was use of stored manure as a fertiliser, while the least expensive system was dumping the vermicompost and fresh manure. However, as discussed earlier, this system is associated with high environmental pollution, making it undesirable. Komakech et al. (2014a) report that 60% of animal manure generated in Kampala is dumped. From Table 8, it appears that one reason for this is the high costs associated with its use as a fertiliser (transport and spreading costs). The cost per kg of nitrogen in vermicompost used as a fertiliser was lower than for fresh or stored manure (Table 8). It should be emphasised that the vermicompost costs presented in Table 8 do not factor in other benefits such as the production of protein. According to Lalander et al. (2015), when the benefits of the vermicompost system (USD 5.92, 2015 price) are included, its new cost becomes USD −0.59 per kg N, representing a net saving. In addition, the N it contains is mainly in the form of nitrates, a more readily available form of nitrogen (Atiyeh et al., 2000), thus making it even more desirable compared with fresh or stored manure. Hence, vermicomposting is probably a more suitable method that ought to be explored in order to reduce the widespread dumping of animal manure in SSA cities such as Kampala.

4. Conclusions

This study measured emissions from a small-scale vermicompost unit located in Kampala, Uganda. The emission factors found for the vermicompost unit were 10.8, 62.3 and 12.8 g Mg^{−1} bio-waste for methane, nitrous oxide and ammonia, respectively. The high nitrous oxide emissions observed were probably because of the high worm density in the unit. In addition, it was established that for the different manure management systems currently in use in SSA cities such as Kampala, the baseline system of dumping the animal waste was the worst performing, especially as regards eutrophication. It is probable that the practice of dumping animal manure is one of the causes of the high levels of eutrophication being experienced in sections of Lake Victoria, bordering Kampala. The high cost of using animal manure as fertiliser could be the reason for dumping animal manure. However, vermicomposting of the animal manure reduces the fertiliser cost, in addition to providing other benefits. It is therefore important for the

authorities to promote vermicomposting in order to reduce dumping of animal manure.

Acknowledgements

We would like to thank the Swedish International Development Cooperation Agency (SIDA) and the Department of Water and Sanitation in developing countries (SANDEC), Swiss Federal Institute of Aquatic Sciences and Technology (EAWAG) for initiating the collaboration and sponsoring this research. We also thank the Department of Agricultural and Biosystems Engineering, Makerere University for providing us with laboratory space and equipment for the experiment.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jenvman.2016.06.028>.

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